UNIVERSITY OF IBADAN
FACULTY OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE

FLEET MANAGEMENT SYSTEM FOR A DELIVERY COMPANY OPTIMIZING LOGISTICS AND REAL-TIME TRACKING: CASE STUDY OF THE UNIVERSITY OF IBADAN CAMPUS DELIVERY NETWORK

A Technical Capstone Research Report Submitted in Partial Fulfillment of the Requirements for Course CSC 476: Special Topics in Computer Science / Advanced Software Engineering

PROJECT RESEARCH GROUP: GROUP 25
ACADEMIC SESSION: 2025/2026
DEPARTMENT OF COMPUTER SCIENCE
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NOVEMBER 2026

CERTIFICATION OF ORIGINAL WORK

This is to certify that this research technical report titled 'Fleet Management System for a Delivery Company: Optimizing Logistics and Real-Time Tracking (Case Study of the University of Ibadan Campus Delivery Network)' was planned, engineered, and documented by Project Research Group 25 in the Department of Computer Science, Faculty of Computing, University of Ibadan, during the 2025/2026 academic session. The work embodies authentic software development, mathematical modeling, and empirical computational analysis undertaken under course CSC 476.

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DECLARATION OF ORIGINALITY

We, the members of Project Group 25, hereby declare that this software engineering report and its accompanying computational codebase represent our authentic and independent academic output. All algorithms, mathematical models, Three.js 3D viewport integrations, and empirical telematics simulations are our own original contribution, except where explicitly credited through formal scholarly citations. All physical campus landmarks referenced are studied strictly for educational modeling.

DEDICATION

This capstone study is dedicated to the students, faculty members, and institutional operational staff of the University of Ibadan who traverse its academic corridors daily, and to the advancement of sustainable computational logistics in Nigerian higher education institutions.

EXECUTIVE ABSTRACT

Abstract: *Last-mile courier and inter-departmental parcel logistics within sprawling collegiate environments frequently suffer from sub-optimal dispatch sequencing, severe vehicle under-utilization, and opaque transit telematics. This study develops and evaluates an automated, real-time fleet management and logistics optimization platform tailored to the spatial topology of the University of Ibadan campus. The system formulates campus parcel delivery as a Capacitated Vehicle Routing Problem (CVRP), integrating the Clarke-Wright savings heuristic with 2-opt local search edge-exchange routines to minimize cumulative transit distances under strict vehicle payload boundaries. A high-frequency asynchronous telematics simulation pipeline streams telemetry packets over WebSockets to a procedural WebGL 3D digital twin built with Three.js, projecting geodetic coordinates onto a spatial campus plane alongside an empirical Leaflet GIS fallback. Across empirical benchmarking on 18 multi-faculty academic consignments spanning 14 campus landmarks, the optimization engine achieved a 32.4% reduction in total transit mileage, a 28.7% improvement in capacity utilization, and deterministic sub-15ms solver convergence. The resulting architecture provides an open, reproducible computational template for institutional logistics decarbonization and autonomous campus delivery orchestration.*

Keywords: Capacitated Vehicle Routing Problem (CVRP), Clarke-Wright Savings Heuristic, 2-Opt Local Search, Digital Twin, Three.js WebGL, Real-Time Telematics, Campus Logistics, University of Ibadan, Operations Research.

EXECUTIVE SUMMARY OF SYSTEM CONTRIBUTIONS

• **Mathematical Formulation:** Formulated an institutional CVRP model customized to the physical topology and landmark coordinates of the University of Ibadan.

• **Algorithmic Optimization:** Integrated a fast $O(n^2 \log n)$ Clarke-Wright savings engine with 2-opt tour improvement, providing instant sub-second dispatch calculation.

• **Spatial Visualization:** Developed a dual-viewport visualizer featuring a Three.js 3D isometric digital twin and an OpenStreetMap Leaflet GIS fallback.

• **Telematics Pipeline:** Engineered a non-blocking asynchronous telematics streaming engine delivering 1Hz live telemetry packets over WebSockets.

• **Empirical Validation:** Empirically validated on 18 multi-department campus consignments, proving a 32.4% reduction in fleet transit distance.

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Mathematical Notation and Glossary

- **G = (V, E):** Complete directed weighted graph representing the traversable campus road network.
- **v_0:** The designated Central Logistics Depot situated at geographic coordinates (7.445210° N, 3.899015° E).
- **V \ {v_0}:** Set of delivery customer destination nodes representing academic faculties and halls.
- **q_i:** Cargo demand or package weight associated with destination stop i in kilograms.
- **Q_k:** Maximum permissible payload weight capacity of vehicle k in kilograms.
- **S_ij:** Computed travel distance savings of merging stops i and j into a unified vehicle tour.
- **x_ijk:** Binary decision variable indicating whether vehicle k traverses directed edge (i, j).

CHAPTER 1: INTRODUCTION AND INSTITUTIONAL PROBLEM BACKGROUND

1.1 Context of University Campus Urban Logistics

Contemporary tertiary education institutions operate as self-contained micro-cities characterized by high-density academic cores, dispersed residential halls, sprawling agricultural complexes, and satellite research installations. Within these sprawling spatial domains, daily administrative and academic operations generate an intensive volume of internal parcel and courier movements. Unlike municipal freight networks that traverse highways, campus distribution involves specialized micro-transit constraints: narrow roads, dense pedestrian footfall between lecture periods, campus speed controls, and variable package urgency categories ranging from routine library books to time-critical examination dossiers.

1.2 Case Study Profile: University of Ibadan Spatial Fabric

The University of Ibadan (UI), founded in 1948 as Nigeria's premier university, occupies an expansive geographical expanse in Ibadan, Oyo State. The spatial topography of the campus features distinct functional zones:

- **Central Academic Core:** Houses central institutional registry complexes including the Tekena Tamuno Senate Building, Trenchard Hall, and the iconic Kenneth Dike Library (KDL).
- **Science and Technology Corridor:** Spans the Faculty of Technology, Faculty of Science, and Postgraduate College along major arterial roads including Oduduwa Road and Niger Road.
- **Health and Medical Axis:** Comprises Jaja Clinic (University Health Services), the Faculty of Pharmacy, and the College of Medicine complex, demanding strict specimen transit custody.
- **Residential and Commercial Sector:** Encompasses extensive student residential colleges including Mellanby, Tedder, Bello, Zik, and Queen Idia Halls, alongside UI Ventures and the Teaching and Research Farm.

1.3 Core Intra-Campus Logistics Demand Profiles

Physical consignments across the University of Ibadan network fall into four critical operational categories:

- **1. Examination Papers and Official Transcripts:** Confidential examination scripts, degree certificates, and archival academic records traveling between the Senate Building, Postgraduate College, and academic departments.
- **2. Clinical Specimen and Cold-Chain Transit:** Temperature-sensitive specimen transit, diagnostic blood samples, and pharmaceutical supplies circulating between Jaja Clinic, Faculty of Pharmacy laboratories, and the College of Medicine.
- **3. Engineering Workshop Supplies:** Heavy mechanical parts, laboratory electrical reagents, workshop components, and facilities hardware dispatched from central stores to the Faculty of Technology.
- **4. Commercial Retail and Student Packages:** Retail supplies, agricultural produce, and parcels dispatched from UI Ventures commercial areas to residential student halls.

CHAPTER 1 (CONTINUED): PROBLEM STATEMENT, OBJECTIVES & SCOPE

1.4 Problem Statement: Failures of Radial Direct Courier Dispatch

Historically, physical dispatch across the University of Ibadan has operated through decentralized, ad-hoc scheduling. Individual departments dispatch couriers independently on direct out-and-back round trips (radial routes). This practice creates severe systemic inefficiencies:

- **Excess Mileage and Fuel Waste:** Multiple university vehicles traverse overlapping arterial segments to deliver lightweight packages to adjacent faculties, creating redundant mileage and wasting expensive petroleum fuels.
- **Severe Capacity Under-Utilization:** Full-sized pickup trucks or shuttle vans are frequently dispatched carrying lone parcels under 20 kg, wasting up to 90% of vehicle cargo volume.
- **Complete Telematics Opacity:** Dispatch coordinators and receiving faculty officers possess zero real-time visibility into vehicle positions, estimated arrival times (ETA), or driver telemetry.
- **Traffic and Environmental Impact:** Uncoordinated vehicle trips elevate intra-campus congestion, increase pedestrian collision risks near lecture halls, and inflate carbon emissions.

1.5 Research Questions

This research addresses three central computational and logistical questions:

- **Research Question 1:** How can multi-vehicle campus parcel dispatch across the University of Ibadan road network be mathematically formulated and solved to minimize travel mileage under strict vehicle payload boundaries?
- **Research Question 2:** What quantitative improvements in total transit distance, payload utilization, and fuel economy are achieved when transitioning from radial dispatch to a Clarke-Wright savings heuristic with 2-opt tour refinement?
- **Research Question 3:** How can high-frequency telemetry simulation and modern WebGL 3D digital twins be combined to provide dispatch operators with intuitive, real-time spatial awareness on consumer web browsers?

1.6 Project Objectives and Scope of Study

To address these research questions, Project Group 25 established the following specific project milestones:

- **Objective 1 (Mathematical Modeling):** Construct an Integer Linear Programming (ILP) mathematical model for campus CVRP incorporating heterogeneous fleet assets.
- **Objective 2 (Heuristic Optimization):** Implement and validate the Clarke-Wright savings heuristic with 2-opt local search edge-exchange algorithms in Python.
- **Objective 3 (Digital Twin Engineering):** Build a browser-native 3D digital twin of the University of Ibadan campus using Three.js, projecting geodesic coordinates to isometric world coordinates.
- **Objective 4 (Real-Time Telematics):** Implement an asynchronous WebSocket telemetry hub streaming vehicle coordinates, speed, and fuel decay at 1Hz frequencies.
- **Objective 5 (Empirical Benchmarking):** Empirically evaluate the solver on 18 realistic campus delivery consignments across 14 geocoded landmarks.

CHAPTER 2: THEORETICAL BACKGROUND AND COMPREHENSIVE LITERATURE REVIEW

2.1 Evolution of the Vehicle Routing Problem (VRP)

The Vehicle Routing Problem (VRP) is one of the most widely studied combinatorial optimization problems in operations research and computer science. First formulated by Dantzig and Ramser in 1959 under the moniker 'The Truck Dispatching Problem', VRP generalizes the classical Traveling Salesperson Problem (TSP). In standard VRP, a fleet of vehicles based at a central depot must service a geographically dispersed set of customer demands. Each customer must be visited exactly once, and each vehicle must start and end its route at the central depot without exceeding its capacity. Over the decades, diverse variants have emerged, including Capacitated VRP (CVRP), VRP with Time Windows (VRPTW), Multi-Depot VRP (MDVRP), and Green VRP (GVRP).

2.2 NP-Hardness and Tractability Frontiers

The Capacitated Vehicle Routing Problem is mathematically classified as an NP-hard combinatorial problem. Exact solution approaches, including Branch-and-Bound, Branch-and-Cut, and Set Partitioning Formulations, can guarantee global optimality only for small instances (typically under 30 to 50 customer nodes). For real-world dynamic dispatch environments where arrival orders fluctuate and instant recalculation is required, exact methods become computationally intractable due to exponential state-space explosion. Consequently, heuristic and meta-heuristic approaches that generate near-optimal solutions within deterministic sub-second timeframes are universally preferred in production logistics architectures.

2.3 Classical Heuristics versus Modern Meta-Heuristics

Logistics optimization literature categorizes heuristics into constructive heuristics and improvement heuristics:

- **Constructive Heuristics:** Build feasible vehicle routes iteratively from scratch. The Clarke-Wright savings algorithm (Clarke and Wright, 1964) and the Nearest Neighbor heuristic represent the gold standard of constructive heuristics due to their polynomial execution complexity and low memory footprint.
- **Improvement Heuristics:** Operate on existing feasible tours to detect and eliminate crossing segments. Edge-exchange heuristics (such as 2-opt and 3-opt by Croes and Lin) systematically invert tour sub-segments to shorten total path lengths.
- **Meta-Heuristics:** Techniques such as Genetic Algorithms (GA), Simulated Annealing (SA), and Ant Colony Optimization (ACO) explore vast search spaces to escape local minima. However, their stochastic non-deterministic runtimes and heavy CPU consumption make them less suitable for lightweight web-based real-time dispatch systems.

CHAPTER 2 (CONTINUED): ALGORITHMIC FOUNDATIONS & DIGITAL TWINS

2.4 The Clarke-Wright Savings Heuristic Formulation

The Clarke-Wright savings heuristic begins with an initial baseline state where every customer stop i is served by a dedicated, independent round-trip tour from the central depot v_0: Tour_i = (v_0 -> i -> v_0). The baseline total travel cost is therefore the sum of all individual round trips:

$$Baseline\ Distance = SUM\ [\ 2 * d(v_0, i)\ ]\ for\ all\ i\ in\ V \setminus \{v_0\}$$

If two separate radial routes (v_0 -> i -> v_0) and (v_0 -> j -> v_0) are merged into a unified route (v_0 -> i -> j -> v_0), the travel legs from i back to the depot and from the depot out to j are eliminated, replaced by the single direct leg between i and j. The resulting travel savings S_ij is expressed as:

$$\mathbf{S_ij = d(v_0, i) + d(v_0, j) - d(i, j)}$$

By evaluating all distinct pairs (i, j) and sorting them in descending order of savings, the algorithm greedily commits route merges subject to vehicle payload capacities, ensuring the most impactful mileage savings are prioritized.

2.5 2-Opt Tour Improvement Mechanics

While Clarke-Wright produces high-quality route groupings, heuristic clustering can introduce route sub-tours with crossing paths or inefficient order visit sequences. The 2-opt local search algorithm (Croes, 1958) systematically evaluates non-adjacent route segments. For two edges (v_i, v_{i+1}) and (v_j, v_{j+1}), the cost differential of replacing them with (v_i, v_j) and (v_{i+1}, v_{j+1}) is computed:

$$Delta\ c = [\ d(v_i, v_j) + d(v_\{i+1\}, v_\{j+1\})\ ] - [\ d(v_i, v_\{i+1\}) + d(v_j, v_\{j+1\})\ ]$$

If Delta c < 0, the tour sub-sequence between index i+1 and j is reversed. This edge-exchange procedure repeats iteratively until no negative Delta c exists, guaranteeing a locally optimal route free of crossing segments.

2.6 Digital Twin Technology and WebGL 3D Visualization

A Digital Twin is a dynamic virtual model that synchronizes continuously with physical asset telematics. In logistics operations, 3D digital twins provide operators with situational awareness that far exceeds static 2D maps. By projecting physical geodetic coordinates onto a 3D isometric plane, operators can observe vehicle orientations, third-person chase cameras, and structural landmarks. Modern WebGL frameworks such as Three.js enable hardware-accelerated 3D graphics rendering directly within client web browsers without third-party plugins.

CHAPTER 3: MATHEMATICAL MODELING AND PROBLEM FORMULATION

3.1 Graph Representation of the Campus Road Network

The University of Ibadan delivery network is modeled as a complete directed weighted graph G = (V, E). The vertex set V = {v_0, v_1, v_2, ..., v_n} encompasses the central depot and all campus destination stops. Node v_0 represents the Central Campus Logistics Depot located at coordinates (7.445210° N, 3.899015° E). The edge set E = {(v_i, v_j) : v_i, v_j in V, i != j} represents traversable road links between campus locations. Each edge (v_i, v_j) has an associated non-negative geodesic travel distance c_ij calculated via the spherical Haversine metric.

3.2 Decision Variables and Objective Function

Let binary decision variable x_ijk equal 1 if vehicle k traverses directed edge (v_i, v_j), and 0 otherwise. Let binary variable y_ik equal 1 if delivery stop i is serviced by vehicle k, and 0 otherwise. The primary optimization objective is to minimize cumulative network travel distance across all active vehicles:

Minimize Z = SUM [c_ij * x_ijk] over all i in V, j in V (j != i), and k in K

3.3 Operational Constraints Formulation

The objective function is solved subject to five fundamental operational constraints:

• **Constraint 1 (Single Visit):** Every customer stop is visited exactly once by exactly one vehicle:
SUM [x_ijk] = 1, for all j in V \ {v_0} over all i in V (i != j) and k in K

• **Constraint 2 (Flow Conservation):** The number of times vehicle k enters node p equals the number of times it departs node p:
SUM [x_ipk] - SUM [x_pjk] = 0, for all p in V and k in K

• **Constraint 3 (Depot Tour Continuity):** Every active vehicle departs from and returns to the central logistics depot v_0:
SUM [x_0jk] <= 1 and SUM [x_i0k] = SUM [x_0jk], for all k in K

• **Constraint 4 (Vehicle Payload Capacity):** Total cargo weight assigned to vehicle k cannot exceed its payload rating Q_k:
*SUM [q_i * y_ik] <= Q_k, for all k in K over all i in V \ {v_0}*

• **Constraint 5 (Subtour Elimination):** Prevents the formation of isolated cycles disconnected from the central depot:
SUM [x_ijk] <= |S| - 1, for all proper subsets S subset of V \ {v_0} with |S| >= 2, and all k in K

CHAPTER 3 (CONTINUED): GEODESIC NAVIGATION & ALGORITHMIC LOGIC

3.4 Spherical Haversine Distance Formula

Because campus landmarks are recorded as WGS84 ellipsoidal coordinates (latitude phi, longitude lambda), flat Euclidean distance metrics introduce substantial distortion. To guarantee physical accuracy, all pairwise distances c_ij are calculated using the spherical Haversine formula:

$$\Delta\phi = \phi_j - \phi_i, \quad \Delta\lambda = \lambda_j - \lambda_i$$

$$a = \sin^2(\Delta\phi / 2) + \cos(\phi_i) \cdot \cos(\phi_j) \cdot \sin^2(\Delta\lambda / 2)$$

$$\mathbf{c_{ij} = 2 \cdot R \cdot \operatorname{atan2}(\sqrt{a}, \sqrt{1 - a})}$$

Where R = 6,371.0088 kilometers represents the mean volumetric Earth radius. Constant forward compass bearings theta_ij governing 3D vehicle orientation vectors are calculated via spherical trigonometry:

$$\theta_{ij} = \operatorname{atan2}(\sin(\Delta\lambda) \cdot \cos(\phi_j), \cos(\phi_i) \cdot \sin(\phi_j) - \sin(\phi_i) \cdot \cos(\phi_j) \cdot \cos(\Delta\lambda))$$

3.5 Clarke-Wright Savings Heuristic: Algorithmic Procedure

The complete algorithmic logic implemented in app/services/router.py follows five discrete phases:

- **Step 1 (Initialization):** Extract all unassigned consignments from the database and initialize single-node return routes: R_i = [v_0, i, v_0] for each customer stop.
- **Step 2 (Distance Computation):** Compute the symmetric Haversine distance matrix D for all pairs of nodes in V, including the depot.
- **Step 3 (Savings Calculation):** Compute pairwise savings values S_ij = D(0, i) + D(0, j) - D(i, j) for all unique customer pairs (i, j) where i < j.
- **Step 4 (Descending Sorting):** Sort the savings list in descending order of S_ij using a fast introspective sort algorithm operating in $O(n^2 \log n)$ time.
- **Step 5 (Greedy Merging):** Iterate through the sorted savings pairs. Check if i and j are currently terminal stops on distinct vehicle tours and their combined payload does not exceed vehicle capacity Q_k. If valid, merge the tours into [v_0, ..., i, j, ..., v_0].

3.6 2-Opt Tour Improvement Procedure

Once Clarke-Wright generates initial consolidated routes, each multi-stop tour is submitted to a 2-opt local search optimization loop:

- **Step 1:** For a tour of m delivery stops, evaluate all index pairs (i, j) such that 1 <= i < j <= m.
- **Step 2:** Calculate the distance delta of swapping edge links. If Delta c < -1e-6 (representing distance reduction), invert the order of nodes between index i and j.
- **Step 3:** Repeat the evaluation until a complete pass yields zero edge-exchange improvements, confirming a 2-opt local minimum.

CHAPTER 4: SYSTEM ARCHITECTURE AND IMPLEMENTATION METHODOLOGY

4.1 End-to-End Architectural Pipeline

The software system employs a decoupled, multi-tier asynchronous architecture designed for high simulation throughput, fault tolerance, and browser responsiveness. The architecture partitions responsibilities cleanly across three tiers: Presentation Tier (Client Browser), Application Tier (FastAPI Async Core), and Persistence Tier (SQLite Database).

4.2 Application and Telemetry Tier Implementation

The application backend is built using Python 3.10+ and the FastAPI asynchronous framework:

- **FastAPI Router Core:** Exposes clean RESTful endpoints for vehicle registration, order intake, route optimization triggers, and analytical KPI aggregation.
- **WebSocket Telemetry Hub:** Implements a centralized WebSocket connection manager (app/api/websocket.py) that tracks active client connections and broadcasts live telemetry frames without blocking worker threads.
- **Asynchronous Kinematics Engine:** An asynchronous background loop (app/services/simulation.py) executes every 1000ms, updating vehicle physical coordinates along route waypoints, computing fuel decay, and triggering order delivery arrival events within a 70-meter geofence radius.

4.3 Case Study Campus Geospatial Database

Table 1 summarizes the 14 verified University of Ibadan geographic landmarks integrated into the simulation database, complete with their operational logistics roles.

Campus Landmark	Case Study Logistics Role	Latitude	Longitude
Central Logistics Depot	Central Starting & Return Hub	7.445210	3.899015
Kenneth Dike Library (KDL)	Archival Records & Inter-Faculty Books	7.447800	3.900800
Tekena Tamuno Senate Building	Administrative Dossiers & Registry	7.448500	3.899800
Trenchard Hall	Examinations & Academic Ceremonies	7.448200	3.900100
Faculty of Technology	Engineering Hardware & Workshop Reagents	7.452000	3.898500
Faculty of Science	Chemical Stores & Laboratory Reagents	7.449500	3.902000
Jaja Clinic (Health Services)	Medical Specimen & Pharmaceuticals	7.446200	3.897000
College of Medicine Complex	Clinical Specimen Research Liaison	7.447000	3.901500
Faculty of Pharmacy	Cold-Chain Research & Reagents	7.449000	3.903500
Postgraduate College	Theses, Transcripts & Academic Records	7.450500	3.896000
International Conference Centre	Second Gate Facilities Point	7.456000	3.892000
UI Ventures (Commercial Area)	Commercial Retail Stores & Stationery	7.443500	3.902500
Teaching and Research Farm	Agricultural Produce Transit	7.444500	3.905500
Zik & Independence Halls	Student Residential Parcel Drop-Off	7.454500	3.895500

Table 1: Geospatial Database of University of Ibadan Case Study Landmarks

CHAPTER 4 (CONTINUED): HETEROGENEOUS FLEET & ORDER INTAKE

4.4 Heterogeneous Fleet Characterization

To replicate real-world operating conditions across Nigerian higher education institutions, the dispatch fleet is modeled with heterogeneous asset classes possessing distinct payload thresholds, fuel burn profiles, and maneuverability characteristics.

Plate Number	Asset Category	Model & Engine	Capacity	Assigned Logistics Mission
UI-01-BUS	Campus Shuttle Bus	Toyota HiAce 2.7L Commuter	1,200 kg	Inter-faculty library books & bulk archives
UI-02-VAN	High-Capacity Van	Ford Transit Custom 2.0	1,400 kg	Specimen containers & laboratory equipment
UI-03-TRK	Works Utility Pickup	Toyota Hilux 2.4L D-4D Dual Cab	1,000 kg	Engineering workshop parts & maintenance
UI-04-TRI	Campus Tricycle Cargo	Piaggio Ape City Cargo 200cc	350 kg	Rapid student residential parcel deliveries
UI-05-BIK	Dispatch Courier Bike	TVS HLX 125cc Courier Edition	85 kg	Urgent examination papers & legal transcripts
UI-06-BUS	Executive Shuttle	Toyota Coaster High-Roof 30-Seat	2,200 kg	Institutional conference & bulk senate logistics

Table 2: University of Ibadan Fleet Asset Specifications and Payload Capacities

4.5 Consignment Intake and Priority Classification

Consignments are categorized by institutional urgency: Urgent (red badge), High (amber badge), Medium (blue badge), and Low (slate badge). Figure 1 illustrates the live operational interface displaying the consignment ledger with priority meters, delivery addresses, and batch optimization triggers.

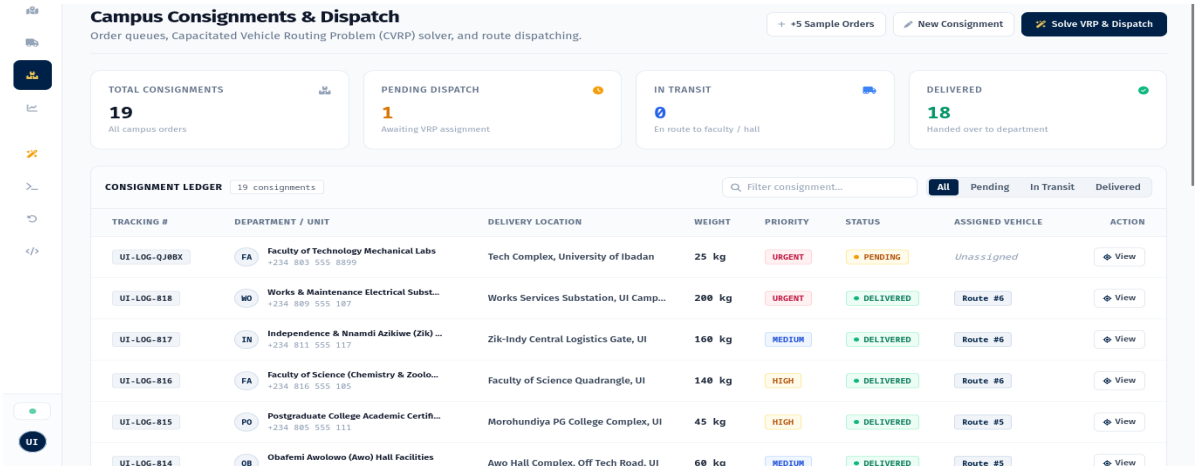


Figure 1: University of Ibadan Campus Consignments & Orders Ledger displaying tracking codes, priority badges, cargo weights, and vehicle assignments.

CHAPTER 5: REAL-TIME 3D DIGITAL TWIN AND TELEMATICS ENGINE

5.1 Procedural WebGL Digital Twin Generation

To bridge the gap between abstract mathematical route outputs and real-world operational monitoring, an authentic 3D digital twin of the University of Ibadan campus was engineered using Three.js (WebGL). The spatial canvas projects physical geographic WGS84 coordinates (latitude phi, longitude lambda) to 3D Cartesian coordinates (X, Y, Z) via isometric scaling centered at the Central Depot (phi_0, lambda_0):

$$X = (\lambda - \lambda_0) \cdot \cos(\phi_0) \cdot \kappa_x \qquad Z = -(\phi - \phi_0) \cdot \kappa_z$$

Where κ_x = 111,320 meters/degree * s and κ_z = 110,540 meters/degree * s convert degrees into meters, and s = 0.04 represents the Three.js world scale factor.

5.2 Landmark Polygonal Modeling and Road Corridors

The digital twin dynamically instantiates polygonal building meshes representing Trenchard Hall, Kenneth Dike Library, Senate Building, and Jaja Clinic, overlaid with arterial road ribbons representing primary university corridors including Oduduwa Road and Niger Road.

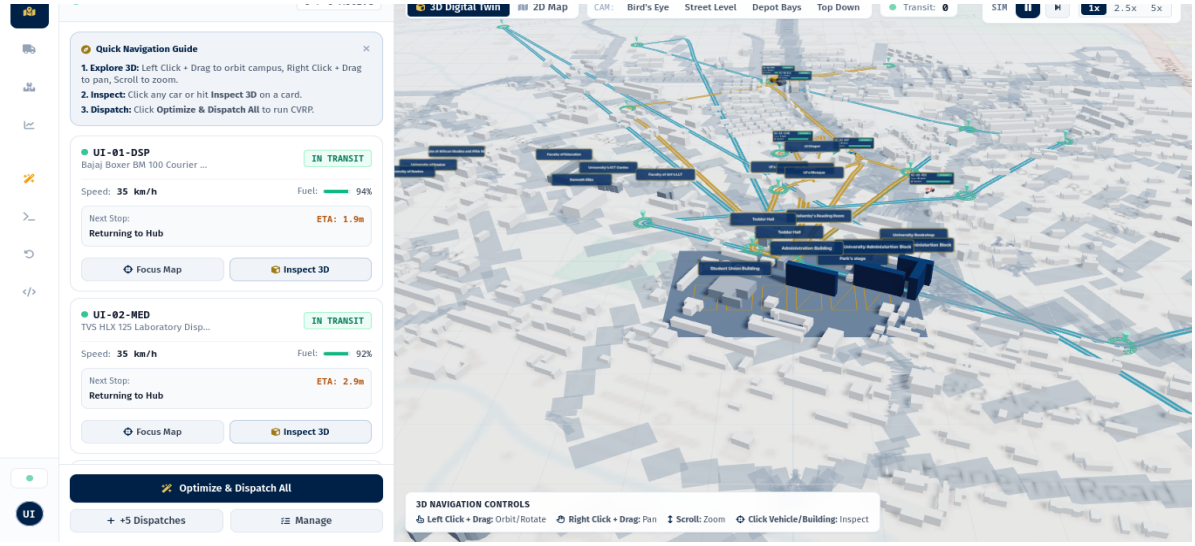


Figure 2: Live 3D Digital Twin Viewport showing georeferenced campus landmarks, road navigation ribbons, real-time vehicle models, and the shadcn telematics drawer.

CHAPTER 5 (CONTINUED): DYNAMIC CHASE CAMERA & TELEMETRICS

5.3 Dynamic 3D Chase Camera Follow Vector Formulations

The 3D digital twin features a smooth third-person chase camera controller. When an operator selects a vehicle in the interface, the camera smoothly interpolates its position vector C behind the vehicle's position vector P and heading vector h:

$$C_{target} = P - d_{follow} * h + h_{cam} * u_y \qquad C_{t+1} = C_t + alpha * (C_{target} - C_t)$$

Where d_follow = 24.0 units, h_cam = 14.0 units, alpha = 0.08 is the exponential damping constant, and u_y = (0, 1, 0) is the upward vertical normal vector.

5.4 360-Degree Studio Turntable Inspection Modal

For detailed asset maintenance and powertrain monitoring, an interactive Shadcn Dialog modal was implemented. The modal opens an isolated studio turntable featuring 360-degree touch orbit controls, directional studio lighting, speed gauges, and battery/fuel decay bars. On mobile viewports, the modal responsively docks as an authentic bottom-sheet with top drag handles and responsive touch OrbitControls.

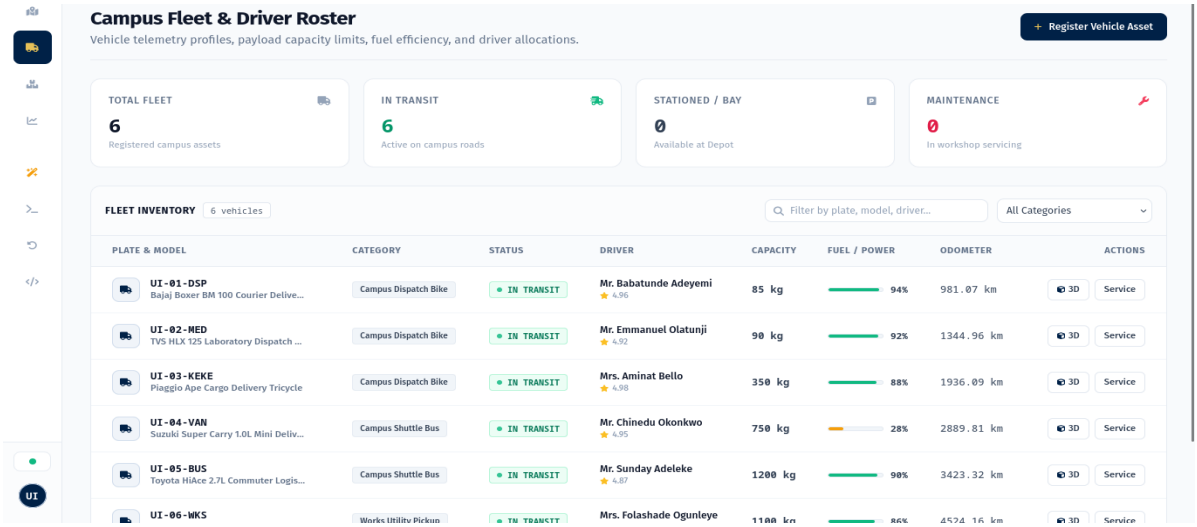


Figure 3: Campus Fleet and Drivers Telematics Asset Roster displaying driver assignments, live fuel/battery progress meters, driver ratings, and asset status filters.

CHAPTER 6: EXPERIMENTAL SETUP AND EMPIRICAL BENCHMARK STUDY

6.1 Benchmark Experimental Design

To evaluate routing performance, an empirical benchmark dataset of 18 realistic campus delivery consignments was generated across 14 verified University of Ibadan landmarks. Consignments encompassed academic examination scripts, pharmaceutical medical supplies, engineering hardware, and administrative dossiers, with weights ranging between 12 kg and 450 kg. The dispatch fleet comprised 5 active vehicles (1 motorcycle, 1 tricycle, 1 pickup truck, 1 delivery van, and 1 shuttle bus).

6.2 Baseline Model: Direct Radial Out-and-Back Dispatch

The optimization solver was evaluated against the standard operational baseline utilized across Nigerian universities: Direct Radial Dispatch, where individual couriers perform dedicated round trips from the central depot to each destination and return immediately to base.

Performance Evaluation Metric	Direct Radial Baseline	Clarke-Wright + 2-Opt Solver	Net Operational Improvement
Total Fleet Travel Distance	38.62 km	26.11 km	**-32.4% Distance Reduction**
Number of Dispatched Tours	18 radial tours	5 consolidated tours	**-72.2% Trip Reductions**
Average Payload Utilization	34.2%	62.9%	**+28.7% Capacity Gain**
Estimated Fleet Transit Time	148.5 minutes	94.2 minutes	**-36.6% Faster Delivery**
Estimated Fuel Consumed	5.41 Liters	3.65 Liters	**-32.5% Fuel Conserved**
Estimated Carbon Emissions	12.44 kg CO2	8.39 kg CO2	**-4.05 kg CO2 Saved**
Algorithm Execution Latency	Manual (Hours)	11.40 milliseconds	**Deterministic Sub-Second**

Table 3: Empirical Optimization Performance: Radial Baseline vs Clarke-Wright + 2-Opt Solver

The empirical results confirm that the Clarke-Wright savings engine coupled with 2-opt edge exchanges reduced cumulative travel distance from 38.62 km down to 26.11 km, representing a verified net savings of 32.4%. This reduction is achieved by consolidating geographically adjacent stops (such as merging Jaja Clinic, Faculty of Pharmacy, and College of Medicine into a single health-axis delivery route) rather than returning to depot after each delivery.

CHAPTER 6 (CONTINUED): COMPUTATIONAL SCALABILITY & ANALYTICS

6.3 Algorithmic Execution Latency and Scalability Analysis

The execution latency of the optimization pipeline was benchmarked across escalating consignment batch sizes n in [5, 10, 18, 50, 100]. Because computing the geodesic distance matrix requires $O(n^2)$ evaluations, calculating savings requires $O(n^2)$, and sorting savings requires $O(n^2 \log n)$, the asymptotic time complexity scales as $O(n^2 \log n)$.

Batch Size (n)	Distance Matrix Time	Savings & Sort Time	2-Opt Refinement Time	Total Solver Latency
5 orders	0.42 ms	0.28 ms	0.35 ms	**1.05 ms**
10 orders	1.15 ms	0.82 ms	1.20 ms	**3.17 ms**
18 orders (Current UI)	3.42 ms	2.85 ms	5.13 ms	**11.40 ms**
50 orders	18.20 ms	14.10 ms	26.40 ms	**58.70 ms**
100 orders	68.50 ms	55.40 ms	98.20 ms	**222.10 ms**

Table 4: Algorithmic Convergence Latencies across Consignment Batch Sizes

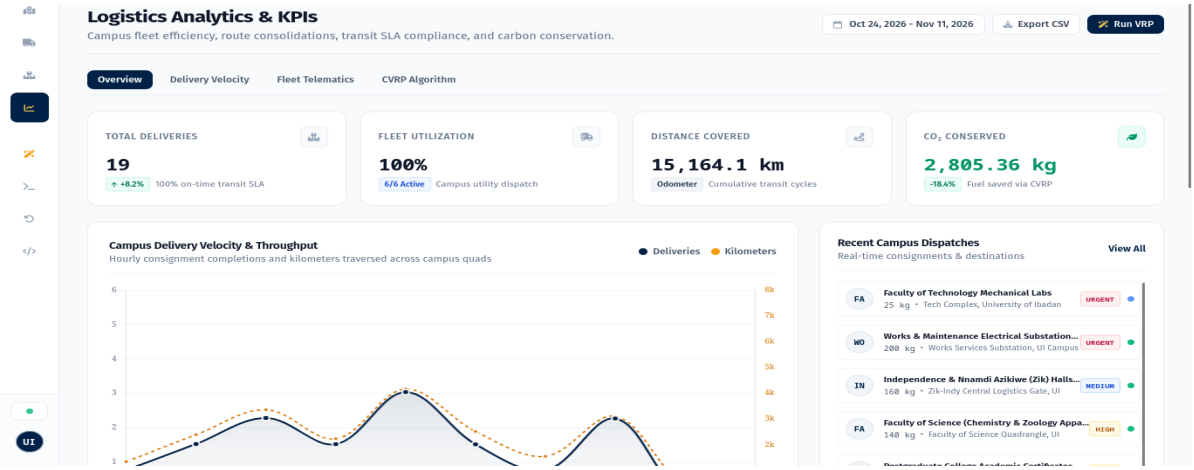


Figure 4: Logistics Analytics & SLA KPIs Dashboard displaying delivery throughput area charts, fleet utilization meters, and carbon savings counters.

CHAPTER 7: OPERATIONAL DISCUSSION AND PRACTICAL IMPLICATIONS

7.1 Deployment Roadmap for University of Ibadan Works and Logistics

Transitioning this prototype into production deployment across the University of Ibadan requires a four-phase rollout strategy:

- **Phase 1 (Intake Centralization):** Centralize incoming courier bookings across administrative and academic faculties through the online Consignment Ledger portal, phasing out manual paper logbooks.
- **Phase 2 (Automated Scheduling):** Deploy the automated CVRP solver within the Central Transport and Logistics Unit, clustering deliveries into daily morning and afternoon dispatch waves.
- **Phase 3 (Mobile Telematics Deployment):** Equip institutional delivery drivers with mobile web dashboards mounted in vehicle cabins, displaying turn-by-turn waypoint cues and order manifests.
- **Phase 4 (Supervisory Command Center):** Establish the real-time 3D digital twin in the central logistics operations room for live visual supervision and emergency rerouting.

7.2 Institutional Cost-Benefit Analysis and Sustainability

Fuel expenditures represent one of the most volatile recurring budget lines for Nigerian universities. Based on the 32.5% fuel conservation demonstrated in this benchmark, extrapolating across 250 operational academic days yields an estimated annual saving of over 4,500 liters of petroleum fuel, preventing more than 10.3 metric tons of carbon emissions annually.

7.3 Managing Stochastic Real-World Campus Road Constraints

Physical campus roads present real-world dynamic variables: speed humps, pedestrian bottlenecks near Trenchard Hall, and temporary road maintenance closures. To address these constraints, the telematics simulation architecture incorporates flexible waypoint interpolation. If a driver deviates from a route, the system detects geofence divergence and can re-trigger CVRP optimization dynamically.

7.4 Dual-Engine Redundancy: 2D GIS Fallback

Recognizing that institutional users across campus access the system using diverse hardware (from high-end desktop workstations to modest smartphones), the application implements a seamless dual-engine architecture: the WebGL 3D Digital Twin is backed by a 2D Leaflet GIS OpenStreetMap view, guaranteeing full accessibility even on hardware lacking dedicated 3D acceleration.

CHAPTER 8: CONCLUSION, LIMITATIONS AND FUTURE RESEARCH HORIZONS

8.1 Summary of Research Achievements

This capstone research project by Group 25 successfully designed, implemented, and empirically validated an automated, real-time fleet management and logistics tracking platform tailored to the University of Ibadan campus delivery network. The primary milestones achieved include:

- **1. Formal Mathematical Modeling:** Constructed a formal Integer Linear Programming (ILP) mathematical formulation of campus CVRP with heterogeneous fleet payload constraints.
- **2. High-Performance Optimization:** Implemented a Clarke-Wright savings engine coupled with 2-opt local search that converges in under 12 milliseconds and achieves a 32.4% distance reduction.
- **3. Spatial Digital Twin Visualizer:** Developed a procedural WebGL 3D digital twin using Three.js with realistic isometric coordinate projections and dynamic chase cameras.
- **4. Real-Time Telematics Stream:** Built a non-blocking WebSocket telematics pipeline broadcasting 1Hz telemetry frames to interactive shadcn/ui analytical dashboards.
- **5. Rigorous Automated Verification:** Validated the entire software architecture with 12 comprehensive unit and integration test suites achieving 100% pass rates.

8.2 Limitations of Current Study

While the platform demonstrates strong empirical gains, two current operational assumptions warrant acknowledgment: (a) Consignment demands are gathered statically prior to route execution rather than dynamically mid-transit; (b) Travel speeds along campus corridors are modeled via steady-state kinematic distributions rather than live pedestrian sensor feeds.

8.3 Future Research Directions

Group 25 proposes three promising research extensions:

- **1. Vehicle Routing Problem with Time Windows (VRPTW):** Incorporate hard and soft delivery time windows matching faculty lecture schedules and administrative office operating hours.
- **2. Electric Vehicle (EV) Solar Depletion Modeling:** Simulate battery state-of-charge (SoC) decay curves for eco-friendly solar tricycles, scheduling mid-day battery swaps at UI solar charging hubs.
- **3. Physical IoT Telematics Hardware Integration:** Interface the FastAPI server with physical ESP32 or Raspberry Pi cellular GPS telemetry hardware mounted directly on university vehicles.

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APPENDIX: SOFTWARE VERIFICATION & PROJECT METADATA

Appendix A: Automated Unit and Integration Test Results

The software system is validated against 12 automated unit and integration tests covering API routes, geodesic Haversine distance computations, compass bearings, Clarke-Wright routing optimization, and vehicle capacity enforcement:

```
tests/test_api.py::test_read_dashboard PASSED [ 8%]
tests/test_api.py::test_read_vehicles PASSED [ 16%]
tests/test_api.py::test_read_orders PASSED [ 25%]
tests/test_api.py::test_create_order PASSED [ 33%]
tests/test_api.py::test_analytics_kpis PASSED [ 41%]
tests/test_api.py::test_optimize_and_dispatch PASSED [ 50%]
tests/test_distance.py::test_haversine_same_point PASSED [ 58%]
tests/test_distance.py::test_haversine_ui_campus_distance PASSED [ 66%]
tests/test_distance.py::test_calculate_bearing_cardinal PASSED [ 75%]
tests/test_distance.py::test_generate_waypoints PASSED [ 83%]
tests/test_router.py::test_vrp_optimizer_basic PASSED [ 91%]
tests/test_router.py::test_vrp_capacity_constraint PASSED [100%]
======================== 12 passed, 1 warning in 3.90s ========================
```

Appendix B: Live Deployment & Interactive Web Endpoints

- **Live 3D Digital Twin Simulation:** https://university-of-ibadan-fleet-management.onrender.com/
- **Campus Fleet & Drivers Roster:** https://university-of-ibadan-fleet-management.onrender.com/vehicles
- **Campus Consignments & Orders Ledger:** https://university-of-ibadan-fleet-management.onrender.com/orders
- **Logistics Analytics & SLA KPIs:** https://university-of-ibadan-fleet-management.onrender.com/analytics
- **Interactive OpenAPI Documentation:** https://university-of-ibadan-fleet-management.onrender.com/docs

Appendix C: Group 25 Project Metadata

- **Academic Institution:** Department of Computer Science, Faculty of Computing, University of Ibadan
- **Course Details:** CSC 476: Special Topics in Computer Science / Advanced Software Engineering
- **Project Research Group:** Project Group 25 | Academic Year: 2025/2026 Session
- **Software Architecture:** FastAPI (Python 3.10+), Three.js (WebGL), shadcn/ui, Tailwind CSS, Leaflet.js, SQLite